

Yit Xiaang Ztang (Yixin Chen)

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Portfolio | GitHub | LinkedIn

EDUCATION

University of Minnesota–Twin Cities, Minneapolis, MN

Sep 2024–Fall 2026

B.S. in Data Science; Minor in Mathematics

GPA: 4.0/4.0

Relevant coursework: Machine Learning, Deep Learning, Optimization, Numerical Analysis, Regression, Natural Language Processing, Software Design

Hefei University of Technology, Hefei, China

Sep 2020–Jan 2024

B.Eng. in Vehicle Engineering; transferred to the University of Minnesota

GPA: 3.75/4.3

PUBLICATIONS

- **Yixin Chen**, B. Fan, J. Li, et al. “An Automated Storage and Retrieval System Optimization with MILP Methods.” *Proceedings of the 5th International Conference on Computer Information Science and Artificial Intelligence (CISAI)*, SPIE, 2023.
- **Yixin Chen**. “Application of Nanomaterials for Improving Zinc-Ion Batteries Performance.” *Journal of Physics: Conference Series*, 2798:012004, 2024.

RESEARCH EXPERIENCE

Randomized Sketching and Sampling for Large-Scale Learning

Jan 2026–Present

Undergraduate Researcher (UROP); Prof. Swati Padmanabhan, University of Minnesota

- Investigate leverage-score sampling, CountSketch, SRHT, subspace embeddings, and sketch-and-solve methods for scalable least-squares and low-rank approximation.
- Analyze norm preservation, approximation guarantees, and Johnson–Lindenstrauss-style embeddings; implement Python/PyTorch benchmarks for runtime, stability, and approximation error.

Reliable PyTorch QP Backend for NCVX and PyGRANSO

Jun 2026–Present

Undergraduate Researcher (UROP); Prof. Ju Sun, University of Minnesota

GitHub

- Developed and validated a dense PyTorch quadratic-programming backend based on OSQP’s operator-splitting structure for sequences of constrained subproblems in PyGRANSO and NCVX.
- Analyzed and integrated reusable KKT/LU factorizations, warm starts, Ruiz scaling, adaptive penalty updates, and polishing, emphasizing conditioning, feasibility, and residual accuracy in the original problem units.
- Benchmarked the backend against OSQP on randomized, ill-conditioned, repeated-solve, and boundary-constrained problems across CPU and CUDA, identifying solver- and backend-level adaptivity as a central performance challenge.

Neural HVAC Cycle Network–Enabled Optimal Control

Dec 2024–Sep 2025

Undergraduate Research Assistant; Prof. Shancong Mou, University of Minnesota

- Developed regression-based predictive models for HVAC thermal-fluid dynamics and 24-hour forecasting, connecting learned system models with optimization-oriented control analysis.
- Performed high-fidelity ANSYS Fluent simulations of heat-exchanger behavior for model validation and closed-loop energy-system analysis.

Inverse Neural Modeling and Magnetic Field Sensing for MIMMS

Jun 2025–Aug 2025

Undergraduate Research Intern; Prof. Hongliang Ren, The Chinese University of Hong Kong

- Implemented multi-head neural models for inverse mapping of magnetically induced metamorphic materials from continuum-simulation data.
- Built visualization workflows for annular magnetic-sensor arrays and distributed force detection on tactile-skin layers.

SELECTED RESEARCH PROJECTS

Weather-Aware Multi-Day Itinerary Optimization and Repair

Jan 2026–Present

Optimization, machine learning, and sequential decision making

GitHub

- Developed a multi-day planning and repair system combining hierarchical mixed-integer optimization, greedy base-lines, and bandit-guided repair under hotel, budget, travel-time, weather, and traveler-preference constraints.
- Built a reproducible pipeline over 228 place candidates with interactive route dashboards and automated validation of feasibility, provenance, route artifacts, and solver status.
- Designed comparative evaluation using utility, feasibility, disruption risk, route preservation, travel time, runtime, and optimality metrics.

Preference Reconstruction and Mechanism Design

Jan 2026–Feb 2026

Mathematical Contest in Modeling; probabilistic modeling, ranking systems, and social choice

- Reconstructed latent fan preferences using Poisson random-utility and softmax models, achieving 72.7% elimination accuracy and 95.6% pairwise agreement.
- Compared three mechanisms across 34 seasons and 335 weeks; designed CHER, a dynamically weighted rule balancing predictive accuracy and fairness.

PDEBench-Lang: Representation Effects in Neural Symbolic PDE Reasoning

Mar 2026–May 2026

- Scientific machine learning, natural language processing, and symbolic reasoning; team project*
- Benchmarked language-model reasoning across natural-language, LaTeX, prefix, and postfix representations of heat, wave, Burgers, Laplace, and advection equations.
 - Evaluated PDE-family classification, structured reasoning chains, representation alignment, and symbolic-pruning fidelity; identified limitations in initial synthetic templates and designed richer benchmark directions.

Representation Transfer and Attention Analysis for Facial Expression Recognition

Mar 2026–May 2026

- Computer vision, transfer learning, and explainable AI; PyTorch and Vision Transformers*
- Compared CNN baselines, ImageNet transfer, face-pretrained models, and Vision Transformers; quantified attention allocation with Grad-CAM and facial-landmark masks.
 - Explored attention-guided objectives that penalize non-semantic background activation to improve interpretability and robustness.

Radar–Camera Fusion for Autonomous Driving

May 2021–Apr 2022

Undergraduate Research Assistant and Team Leader; Prof. Junzhao Jiang, Hefei University of Technology

Developed multimodal perception models with OpenCV, PyTorch, and MMDetection; built calibration and debugging setups for autonomous-driving experiments.

TEACHING

Teaching Assistant, CSCI 2081: Introduction to Software Design

Sep 2025–May 2026

University of Minnesota

Supported labs, office hours, debugging, and Java/OOP instruction; contributed data-science examples and Docker-based laboratory materials.

SELECTED HONORS

Two UROP Awards , University of Minnesota	2026
Dean’s List , University of Minnesota	2024–2026
National Scholarship	2021
Second Prize , HFUT Mathematics Competition	2021

TECHNICAL SKILLS

Programming: Python, C++, Java, R, MATLAB, SQL | **Optimization/ML:** PyTorch, Gurobi, OSQP, XGBoost, NumPy, SciPy, scikit-learn

Research tools: Linux, Git, Docker, Pandas, ANSYS Fluent | **Methods:** optimization, randomized linear algebra, regression, deep learning, time-series modeling